

Huanyu WANG

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Education

Shanghai Jiao Tong University, Shanghai, China Sept. 2022 - Jun. 2026 (Expected)
B.S. in Computer Science & B.S. in Applied Mathematics
GPA: 90.3/100 (Overall)
Courses: Game Theory (97), Operating System (95), Programming Languages and Compilers (95), Data Structure (95), Principles and Methods of Program Design (95), Design and Analysis of Algorithms (95), Program Design Practice (94.5), Machine Learning (93), Computer Vision (93), Computer Networks (93).
High School Affiliated to SJTU, Shanghai, China Sept. 2019 - Jun. 2022
Top 100 scorers in Shanghai College Entrance Exam

Publications

- **Huanyu Wang**, Ziyu Xia, Zhuoming Chen, Beidi Chen.
WWW.Serve: Interconnecting Global LLM Services through Decentralization.
arXiv:2603.20661.
- **Huanyu Wang**, Jushi Kai, Haoli Bai, Lu Hou, Bo Jiang, Ziwei He, Zhouhan Lin.
Fourier Compressor: Frequency-Domain Visual Token Compression for Vision-Language Models.
arXiv:2508.06038.

Research Experiences

Research Assistant, InfiniAI Lab, Carnegie Mellon University Oct. 2025 - Jan. 2026
Advisor: Prof. Beidi Chen.

- Reframed LoRA-based RL training as a **service-oriented system**, modeling each rollout-training step as a schedulable request to enable multi-tenant RL-as-a-Service.
- Designed a **router-based framework** that orchestrates requests across multiple rollout servers and trainers, balancing heterogeneous resource demands and improving overall hardware utilization.
- Leveraged the stateless property of LoRA adapters to enable fire-and-forget scheduling, flexible decoupling of rollout and training, and dynamic participation of training services.

Research Assistant, InfiniAI Lab, Carnegie Mellon University Mar. 2025 - Oct. 2025
Advisor: Prof. Beidi Chen.

- Tackled scalability bottlenecks, scheduling inefficiencies, and privacy concerns in centralized multi-LLM serving systems through a novel decentralized approach.
- Designed *WWW.Serve*, a **fully decentralized framework** for trustless collaboration across heterogeneous LLM servers, featuring autonomous workload balancing, flexible policies, and privacy protection.
- Evaluated the framework under diverse configurations and workloads, demonstrating up to **1.5×** improvement in global SLO attainment and **27.6%** reduction in request latency.

Research Assistant, LUMIA Lab, Shanghai Jiao Tong University Oct. 2024 - Aug. 2025
Advisor: Prof. Zhouhan Lin

- Addressed the challenge of computational redundancy in Vision-Language Models (VLMs), significantly improving the efficiency of inference while maintaining competitive accuracy.
- Developed *Fourier Compressor*, which integrates the **2-dimensional Discrete Cosine Transform** with LLaVA and Qwen-VL architectures, achieving a **93.75%** reduction in vision token count.
- Conducted comprehensive evaluations across multiple benchmarks, maintaining **96.1%** performance while achieving **83.8%** FLOPs reduction, **86.4%** less KV cache usage, and **31.2%** faster inference speed.

Skills

Programming: Python, PyTorch, Transformers, Git, Linux, Hugging Face, C++, LaTeX.
Mathematics: Linear Algebra, Discrete mathematics, Information Theory, Probability & Statistics, ODE.
Language: English - TOEFL: 107 (Speaking 24).